Superconducting Devices

From wastewater treatment to MEG and buried-metal detection

One physics, two device families, three applications

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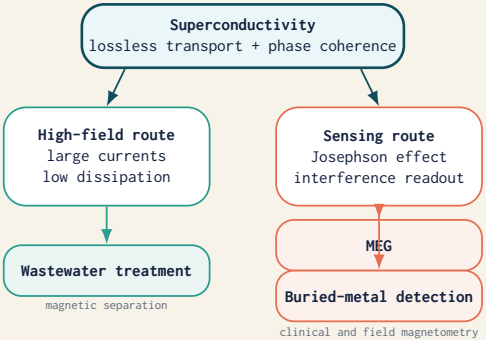
Guiding idea

- Superconductivity becomes a technology platform in **two complementary ways**:
  - it carries very large currents with negligible Joule losses
  - it enables phase-coherent magnetic sensing through the Josephson effect
- This gives two device branches:
  - **superconducting magnets** for strong-field action
  - **SQUIDs** for ultra-sensitive readout
- The talk follows one claim: **shared physics, different engineering targets**.

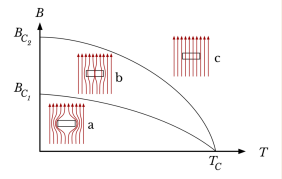


What makes a superconductor technologically useful?

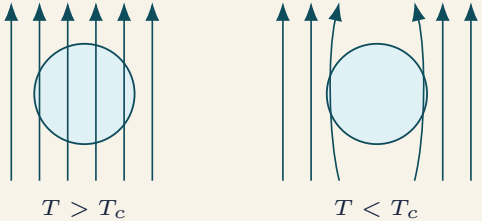
- **Zero resistance** allows persistent currents without macroscopic dissipation.
- **Meissner expulsion** shows that superconductivity is a true thermodynamic phase, not just perfect conductivity.
- In **type-II materials**, flux enters through quantized vortices between H_{c1} and H_{c2}.
- This mixed state is what makes high-field superconducting technology practical.

$$\nabla^2 \mathbf{B} = \frac{\mathbf{B}}{\lambda_L^2} \qquad \Phi_0 = \frac{h}{2e}$$

Takeaway: The same material platform supports both strong magnetic action and precise magnetic control.



Type-I screening, vortex mixed state, and field penetration above H_{c2}.



Meissner effect: below T_c the magnetic field is expelled.

Macroscopic phase and the Josephson effect

- The superconducting state is described by a macroscopic wavefunction

$$\psi(\mathbf{r}) = |\psi(\mathbf{r})| e^{i\theta(\mathbf{r})}$$

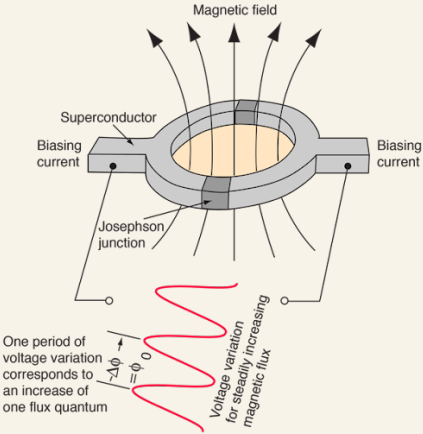
- **Key idea: the phase θ is physical.**
- A phase difference between two superconductors generates a supercurrent:

$$I_s = I_c \sin\varphi$$

- An applied voltage changes the phase in time:

$$\frac{d\varphi}{dt} = \frac{2e}{\hbar} V$$

Phase difference $\longrightarrow$ current voltage $\longrightarrow$ phase evolution



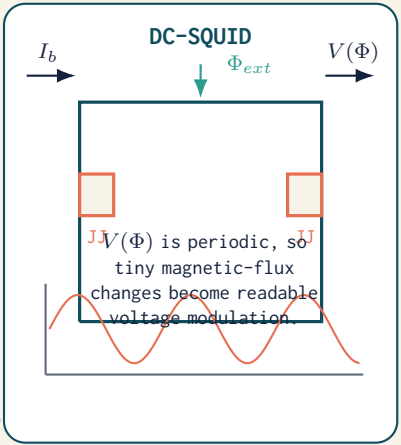
Josephson junctions are the key element for superconducting sensing.

SQUID: flux-to-voltage transducer

- A DC-SQUID is a superconducting loop with two Josephson junctions.
- Magnetic flux changes the phase around the loop.
- This modifies the maximum supercurrent the device can carry.
- In first approximation,

$$I_c(\Phi) \simeq 2I_0 \left| \cos\left(\pi \frac{\Phi}{\Phi_0}\right) \right|$$

- Electronics convert this modulation into a measurable voltage signal.



Takeaway:
Magnetic flux is converted into voltage through quantum interference.

From phase to measurement (intuition)

- Magnetic field $\rightarrow$ magnetic flux through the loop
- Flux $\rightarrow$ phase shift in the superconducting wavefunction
- Phase shift $\rightarrow$ change in supercurrent (Josephson effect)
- Supercurrent change $\rightarrow$ voltage modulation (readout electronics)

Big picture

A SQUID does not measure magnetic field directly: it translates it into phase, then into current, then into voltage.

Takeaway: This chain is what makes quantum interference usable as a practical sensor.

Application 1: wastewater treatment by magnetic separation

- Pollutants are first attached to magnetic seeds or flocculants.
- The flow then crosses a high-gradient magnetic separator driven by a superconducting magnet.
- Magnetic force acts on the tagged contaminants, while cleaned water continues downstream.
- Here the key resource is **high field and high gradient**, not quantum interference.

Takeaway: This is the strong-field branch of the story: superconductivity is useful because it generates magnetic action efficiently.

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Why use a superconducting magnet here?

- Resistive magnets can produce strong fields, but continuous operation becomes energetically expensive.
- Superconducting coils enable stable high-field operation with much lower Joule losses.
- That makes large-scale magnetic separation more realistic in industrial settings.
- The engineering question is therefore system integration: cryogenics, throughput, and process cost.

	Resistive	**SC**
High field	possible	practical
Continuous duty	costly	favorable
Power dissipation	high	low in the coil
System constraint	power supply	cryogenics

Message

The superconductor is the actuator, not the sensor.

Application 2: magnetoencephalography with SQUID arrays

- MEG measures external brain magnetic fields that can be as small as 10^{-15} T.
- Whole-head systems use large SQUID arrays inside cryogenic dewars.
- Clinically, MEG is important for pre-surgical epilepsy localization and functional mapping.
- The technological value is extreme sensitivity combined with millisecond time resolution.

Takeaway: In this branch the limiting factor is not force, but the ability to read out extraordinarily weak fields non-invasively.

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Application 3: buried-metal and landmine detection

- A transmitter coil induces eddy currents in the buried metallic target.
- The target then emits a weak secondary magnetic field.
- A SQUID is attractive because low-frequency sensitivity remains extremely strong.
- In field deployment, performance must coexist with shielding limits, motion, and cryogenic constraints.

Takeaway: This is still the sensing branch, but now in a much noisier and less controlled environment than MEG.

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One physics, three engineering targets

Application	Useful quantity	Why superconductivity helps	Main constraint
Wastewater treatment	High field and high gradient	Efficient magnetic separation with reduced coil losses	Cryogenics and plant integration
MEG	Brain fields down to femtotesla scale	Ultra-sensitive flux readout with fast temporal response	Shielding and low-noise environment
Buried-metal detection	Weak secondary low-frequency field	Better low-frequency sensitivity than conventional detectors	Robustness outside controlled settings

Takeaway: The unity comes from superconducting physics; the diversity comes from what the device is optimized to do.

Limits and outlook

Current limits

- Cryogenic infrastructure
- Magnetic shielding and noise control
- System complexity outside the lab
- Cost and maintenance

Where progress matters

- Better HTS materials and devices
- Simpler cryocooler-based operation
- More robust low-noise electronics
- Easier deployment in clinical and field environments

Takeaway: The open challenge is no longer only physical understanding, but engineering simplification and scale-up.

Conclusion

- Superconductivity is useful because it can either **create** magnetic fields efficiently or **detect** magnetic fields with quantum-level sensitivity.
- The theory block leads naturally to devices: Meissner response, phase coherence, Josephson effect, flux quantization, then SQUID operation.
- Water treatment, MEG, and buried-metal detection are not disconnected examples; they are different outcomes of the same underlying physics.

Take-home message:

Superconductivity becomes a technology platform because it supports both magnetic action and magnetic readout.

References I

Course material

- R. Arpaia, Lecture 4 – Superconductivity. Key Phenomena and Concepts.
- R. Arpaia, Lecture 7 – Superconductivity. London Equations (I).
- R. Arpaia, Lecture 15 – The Josephson Effect. Gateway to Superconducting Device Physics (II).
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Applications

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- Li, J. et al., FeOOH and nZVI combined with superconducting high gradient magnetic separation for the remediation of high-arsenic metallurgical wastewater, Separation and Purification Technology 285 (2022) 120372.
- Hari, R. et al., IFCN-endorsed practical guidelines for clinical magnetoencephalography (MEG), Clinical Neurophysiology 129 (2018) 1720–1747.

References II

- Guo, X. et al., Metal detector based on high-T_c RF SQUID, Physica C 378–381 (2002) 1404–1407.